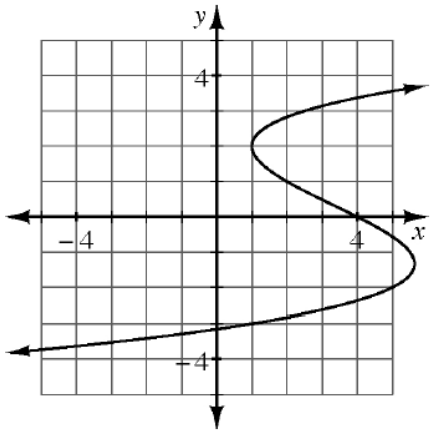
Lesson 1.2.5 — Domain and Range

LAUNCH

1-71



Examine the graph of the relationship shown here. Use it to estimate:

a. y when $x = -4$

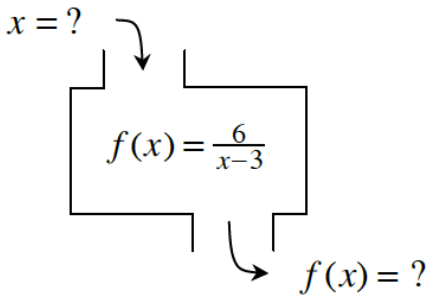
b. y when $x = 1$

c. y when $x = 4$

d. Is this relationship a function? Why or why not?

EXPLORE

1-72

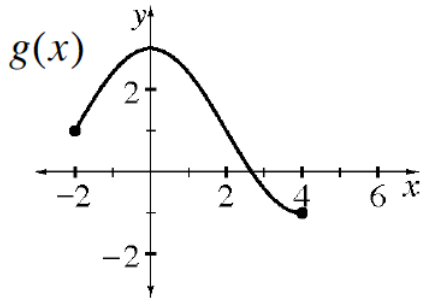


Examine the relation shown here.

1. Find $f(-3)$, $f(0)$, and $f(2)$.
2. Find $f(3)$. What happened?
3. Are there any other inputs that cannot be evaluated by this function? In other words, are there any other values that x cannot be? Explain how you know.
4. The set (collection) of numbers that can be used for $x$in a function and still get an output is called the domain of the function. The domain is a description or list of all the possible x-values for the function. Describe the domain of $f(x) = \frac{6}{x-3}$.
5. What other types of functions have you looked at thus far in this course or in previous courses that have limited domains? How were they limited?

1-73

Now examine $g(x)$ graphed shown here.



1. Is $g(x)$ a function? How can you tell?
2. Which x-values have points on the graph? That is, what is the domain of $g(x)$?
3. What are the possible outputs for $g(x)$? This is called the range of the function.
4. Ricky thinks the range of $g(x)$ is: $-1, 0, 1, 2,$ and 3. Is he correct? Why or why not?
5. What other functions have you worked with previously in this course or previous courses that have limited ranges? How were they limited?

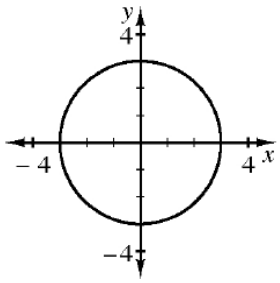
1-74

FINDING DOMAIN AND RANGE

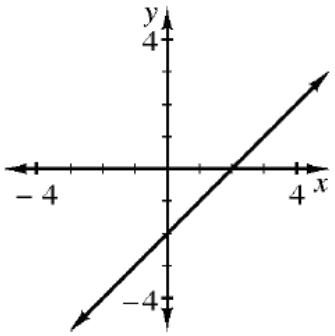
The domain and range are good descriptors of a function because they help you know what numbers can go into and come out of a function. The domain and range can also help you set up useful axes when graphing and help you describe special points on a graph (such as a missing point or the lowest point).

Work with your team to describe in words the domain and range of each relationship below. Then state whether or not the relationship is also a function.

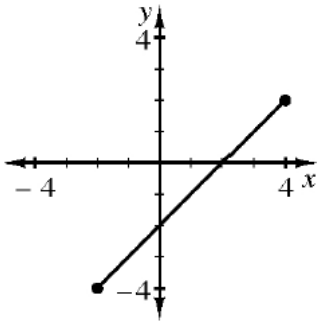
a.



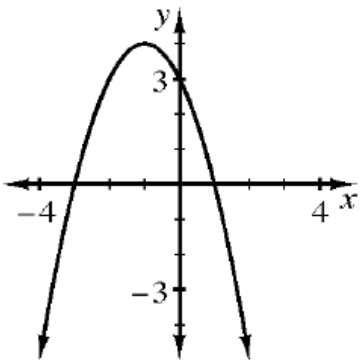
b.



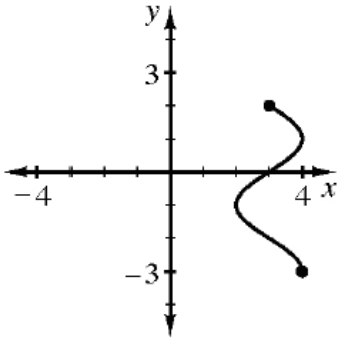
c.



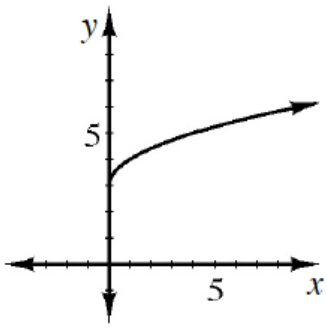
d.



e.



f.



1-75

Chiu loves tables! He has decided to make the table shown here for a function $f(x)$ to help him find its domain and range.

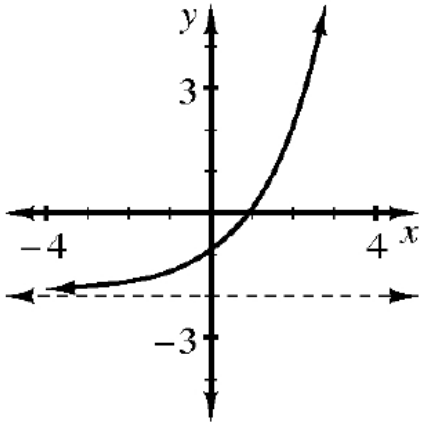
a. From his table, can you tell what the domain of $f(x)$ is? Why or why not?

b. From the table, can you tell the range of $f(x)$? Why or why not?

c. Is using a table an effective way to determine the domain and range of a function?

x	$f(x)$
-3	5
-2	0
-1	-3
0	-4
1	-3
2	0
3	5

1-76



Daniel is thinking about the relation shown here.

a. He noticed that the curve continues shown here and shown here. What is the domain of this function?

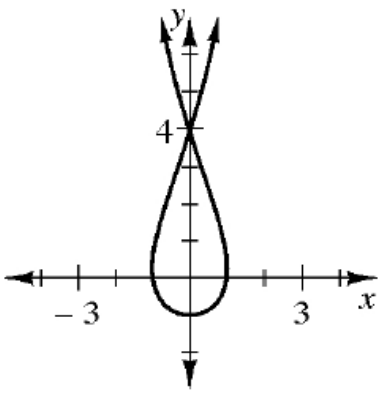
b. He found out that the dotted line is a line that the graph gets closer and closer to as x gets very, very negative. (Another name for this dotted line is asymptote.) How should Daniel describe the range?

REVIEW & PREVIEW

1-78

Which of the relationships below are functions? If a relationship is not a function, give a reason to support your conclusion.

a.



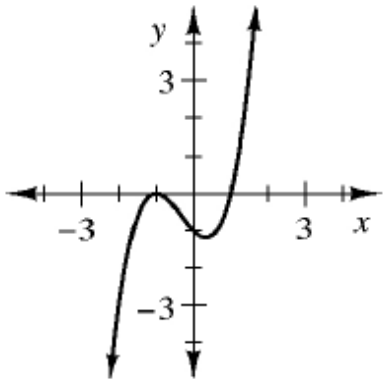
b.

x	y
-3	19
5	19
19	0
0	-3

c.

x	7	-2	0	7	4
y	10	0	10	3	0

d.



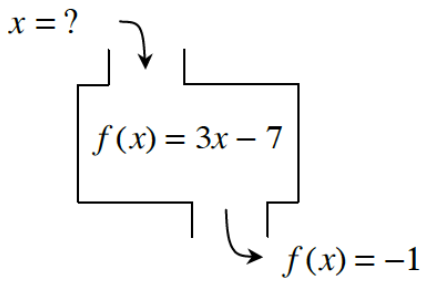
1-79

Find the x- and y-intercepts for the graphs of the relationships in problem 1-78.

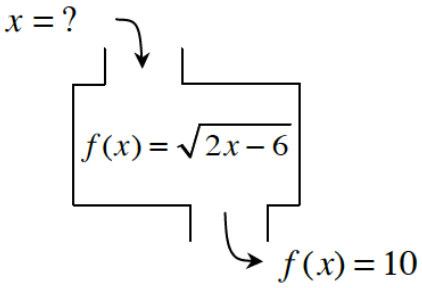
1-80

Find the inputs for the following functions with the given outputs. If there is no possible input for the given output, explain why not.

a.



b.



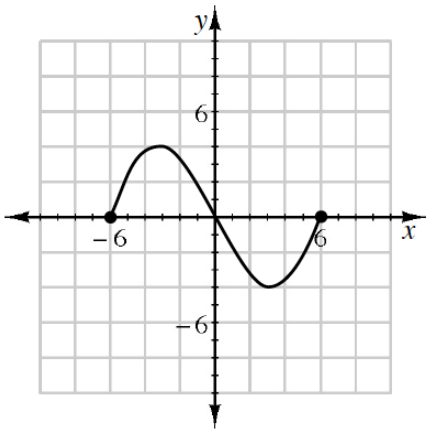
1-81

Use the relationship graphed shown here to answer the questions below.

a. Is the relation a function?

b. What is the domain?

c. What is the range?



1-82

What value(s) of x will make each equation true?

a. $\sqrt[3]{x} = -2$

b. $\sqrt{x} = 12$

c. $|x + 1| = 4$